Persona-Aware Fashion Image Understanding and Caption Generation for Marketing

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Abstract

Personalized marketing in fashion e-commerce requires not only accurate visual understanding of products but also stylistic adaptation of marketing language to different user personas. While recent vision–language models have achieved strong performance in factual image captioning, their outputs are often neutral and lack controllable stylistic variation, which is critical for marketing applications.

In this project, we explore a vision-centric framework for persona-aware fashion caption generation that relies solely on image inputs at inference time. Instead of using CLIP as a direct captioning tool, we employ it as a structured visual reasoning module that decomposes fashion images into interpretable attributes such as category, occasion, color, pattern, and persona-related style cues. These intermediate representations are then reused across multiple caption generation strategies, including a compositional rule-based method, a persona-controlled template method, and a CLIP-guided generative vision–language model.

By comparing these strategies under a unified visual understanding pipeline, our work highlights the trade-offs between faithfulness, controllability, and linguistic fluency in persona-aware marketing caption generation. The results demonstrate that structured visual inference enables transparent and flexible control over marketing language without requiring metadata at test time, offering a practical design for vision-driven personalization systems.

1. Introduction

Fashion e-commerce platforms increasingly rely on visual content to communicate product value and brand identity. Unlike generic object descriptions, fashion marketing captions are expected to convey subtle stylistic cues such as mood, lifestyle, and intended usage, often tailored to different consumer personas. However, most existing image captioning systems focus primarily on factual description and offer limited control over stylistic variation, making them less suitable for marketing-oriented scenarios.

Recent advances in vision–language models (VLMs) have significantly improved the quality of image captioning by leveraging large-scale multimodal pre-training. Models such as BLIP and LLaVA can generate fluent and descriptive captions across diverse visual domains. Despite these successes, their outputs tend to remain neutral in tone and often lack explicit mechanisms for controlling stylistic attributes such as persona or marketing intent. This limitation becomes particularly apparent in fashion applications, where the same product may need to be presented differently depending on the target audience.

In this report, we investigate whether structured visual understanding can serve as a foundation for controllable, persona-aware caption generation in fashion marketing. Rather than relying on end-to-end caption generation alone, we decouple visual understanding from language generation. A pre-trained CLIP model is used to extract interpretable semantic attributes from fashion images, which are then reused as intermediate representations for multiple caption generation strategies. This design allows us to systematically analyze how different language generation approaches behave under the same visual constraints.

The primary contribution of this work lies not in applying CLIP or VLMs individually, but in demonstrating how CLIP can be repurposed as a structured visual reasoning module whose outputs enable transparent and controllable marketing caption generation. By focusing on system design and comparative analysis, this project aims to provide insights into how multimodal AI systems can bridge visual perception and stylistic language generation in practical marketing contexts.

2. System Overview

The overall pipeline separates visual understanding from language generation to enable modular analysis. Given a single fashion product image as input, the system first performs visual attribute inference using CLIP and then generates marketing captions conditioned on the inferred attributes.

The system consists of three stages: (1) CLIP-based visual attribute inference, (2) persona inference via zero-shot similarity to persona descriptors, and (3) caption generation using multiple strategies. By fixing the visual inference stage and varying only the captioning strategy, we isolate the effect of different language generation approaches on caption style, fluency, and faithfulness.

3. Visual Attribute and Persona Inference

We employ a pre-trained CLIP model (ViT-B/32) as the core visual understanding component. CLIP maps images and text into a shared embedding space, allowing semantic similarity to be measured through cosine distance. We design a structured inference process in which the model evaluates similarity between an image embedding and multiple sets of textual candidates representing product category, usage occasion, colors, patterns, and persona-related style cues.

For each image, CLIP similarity is computed against predefined textual candidates. The highest-scoring candidates are selected as inferred attributes, enabling interpretable, reusable intermediate representations. Persona is inferred similarly using a small set of persona descriptors (e.g., minimal, classic, street, romantic). Although these labels are not ground-truth annotations, they provide a meaningful approximation of stylistic intent derived directly from visual features.

4. Caption Generation Strategies

Using the same set of CLIP-inferred attributes, we compare three caption generation strategies with different trade-offs between controllability and expressiveness.

4.1. CLIP-only Compositional Baseline

This strategy constructs captions deterministically by composing inferred attributes into sentence templates. It provides strong faithfulness and reproducibility, but limited linguistic diversity.

4.2. Persona-controlled Template Captioning

This strategy extends the compositional baseline by introducing persona-dependent stylistic modifiers while keeping factual content fixed. It enables tone variation without generative models.

4.3. CLIP-guided Generative Captioning

This strategy uses a generative vision–language model guided by CLIP-inferred attributes embedded in prompts. It produces the most fluent captions but may occasionally drift from inferred attributes.

5. Experiments and Results

We evaluate the generated captions using qualitative inspection, focusing on persona consistency, marketing appropriateness, and potential hallucinations. Across methods, the compositional baseline yields the most faithful but least expressive captions. Persona-controlled templates improve stylistic variation while preserving content stability. CLIP-guided generation achieves the most natural and marketing-like text, with occasional attribute drift.

All experiments are conducted in a vision-only inference setting, where no textual metadata is provided at test time. For each image, visual attributes and persona are inferred using CLIP, and the same inferred attributes are used across all caption generation strategies. This controlled setup allows us to attribute differences in generated captions solely to the captioning mechanism rather than variations in visual understanding.

5.1. Qualitative Comparison of Caption Generation Methods

To analyze behavioral differences among caption generation strategies, we qualitatively compare captions generated from the same fashion image using identical CLIP-inferred visual attributes. By fixing the visual inference stage and varying only the caption generation mechanism, this comparison isolates the effect of language generation strategies on stylistic expression, linguistic fluency, and faithfulness to visual content.

Specifically, for each image, three captions are presented: (1) a deterministic CLIP-only compositional caption, (2) a persona-controlled template-based caption, and (3) a generative caption produced by a vision–language model (LLaVA). This design enables a controlled examination of how different captioning approaches behave under the same visual constraints.

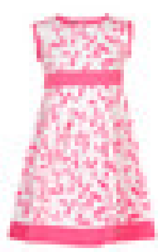
Image	Method / Persona	Generated Caption
	CLIP-only Persona (Minimal) VLM (LLaVA)	Purple jacket with a clean, minimal look, perfectly suited for sportswear. Purple Jacket with a clean, understated look, built for sportswear. A purple and white hooded sweatshirt with the zipper pulled up to its chin.
	CLIP-only Persona (Classic) VLM (LLaVA)	White graphic shirt featuring a distinctive graphic design, perfectly suited for casual everyday wear. White graphic Shirt with a soft, elegant mood, perfect for casual everyday wear. Fall in love with this whimsical print
	CLIP-only Persona (Classic) VLM (LLaVA)	Pink graphic dress featuring a distinctive graphic design, perfectly suited for formal office wear. Pink graphic Dress with a soft, elegant mood, perfect for formal office wear. A pink floral print dress with a matching belt and sash.
	CLIP-only Persona (Classic) VLM (LLaVA)	Grey shoes with a clean, minimal look, perfectly suited for sportswear. Grey Shoes with a timeless, refined style, suited to sportswear. Sleek and stylish athletic footwear for the modern athlete.

Table 1. Qualitative comparison of persona-aware fashion marketing captions. Each row shows one input image and multiple captions generated by different strategies under identical CLIP-inferred visual attributes.

The qualitative results reveal clear differences among caption generation strategies. The CLIP-only compositional captions strictly adhere to inferred visual attributes, ensuring high faithfulness and reproducibility but resulting in relatively rigid and formulaic language. Persona-controlled template captions preserve factual consistency while introducing stylistic modifiers aligned with the inferred persona, demonstrating that tone variation can be achieved without generative models.

In contrast, the generative captions produced by the vision–language model exhibit higher linguistic fluency and stronger marketing appeal. However, these captions occasionally introduce additional descriptive details not explicitly inferred during the visual reasoning stage, such as specific garment components or implied design elements. This behavior highlights a trade-off between expressive richness and strict faithfulness to visual inference.

An additional observation arises in cases where CLIP-inferred attributes differ from human intuition. For example, certain dresses are inferred as suitable for formal office wear despite exhibiting floral patterns or vivid colors. In such cases, all captioning strategies consistently follow the inferred attributes, indicating that errors or biases in visual inference propagate uniformly to downstream caption generation. This underscores the importance of reliable visual reasoning as a foundation for controllable persona-aware captioning.

6. Discussion

The experimental results suggest that persona-aware caption generation should be approached primarily as a system design problem rather than a purely modeling problem. By fixing the CLIP-based visual reasoning stage and varying only the caption generation strategy, we observe that controllability and reliability are more strongly influenced by the structure of visual inference than by the expressive power of the language model itself.

The CLIP-only compositional baseline demonstrates that structured visual inference alone is sufficient to produce faithful and consistent captions. Although the resulting language is rigid, this approach offers complete transparency and predictability, which are valuable properties in commercial settings where incorrect or misleading descriptions can undermine user trust. Importantly, this baseline establishes that persona-aware variation does not inherently require generative models.

Persona-controlled template captioning further illustrates that stylistic differentiation can be achieved by explicitly separating content from tone. Because persona information is injected after visual attributes are fixed, this strategy enables controlled stylistic variation without compromising factual consistency. This observation reinforces the importance of modular system design, where persona is treated as a controllable variable rather than an implicit property learned end-to-end.

In contrast, CLIP-guided generative captioning produces the most fluent and engaging marketing language, closely resembling human-written descriptions. However, qualitative analysis reveals that increased expressiveness comes at the cost of occasional attribute drift and hallucinated details. These deviations highlight a fundamental trade-off between linguistic creativity and strict adherence to visual evidence, suggesting that generative models should be carefully constrained when used in applications requiring high factual reliability.

Another notable finding concerns the propagation of visual inference errors. When CLIP infers attributes that diverge from human intuition—such as categorizing visually decorative garments as formal office wear—all downstream captioning strategies consistently reflect this inference. This behavior underscores the central role of visual reasoning in the overall system and suggests that improvements in caption quality are more effectively achieved by refining visual attribute inference than by increasing language model complexity.

Overall, the discussion emphasizes that persona-aware caption generation is best approached as a system design problem rather than a purely modeling problem. Structured visual reasoning provides a stable foundation for controllable language generation, while different captioning strategies allow designers to balance faithfulness, stylistic variation, and fluency depending on application requirements.

7. Conclusion

This report presented a vision-based framework for persona-aware fashion marketing caption generation that operates solely on image inputs at inference time. By repurposing CLIP as a structured visual reasoning module, the proposed system decouples visual understanding from language generation, enabling transparent, interpretable, and controllable captioning.

Through qualitative comparison of three caption generation strategies—CLIP-only compositional captions, persona-controlled template captions, and CLIP-guided generative captions—we demonstrated that stylistic variation and linguistic fluency are largely determined by the captioning mechanism, while content fidelity is governed by the underlying visual inference. The results highlight clear trade-offs between controllability and expressiveness, showing that increased linguistic naturalness often introduces risks of attribute drift and hallucination.

A key contribution of this work lies in its system-level perspective. Rather than focusing on training or fine-tuning larger models, the project illustrates how careful decomposition of visual reasoning and language generation can yield flexible and reliable persona-aware outputs. This design philosophy is particularly relevant for fashion marketing applications, where transparency, consistency, and stylistic control are as important as fluency.

While this study focuses on qualitative analysis, future work could extend the framework with quantitative evaluation metrics, improved persona modeling, or lightweight fine-tuning methods such as LoRA to better align generative models with structured visual constraints. Nevertheless, the presented results suggest that structured vision-driven personalization offers a practical and interpretable alternative to fully end-to-end caption generation systems.

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